UNIT V : Dynamic Analysis

5.1 Torsion of a Uniform Shaft

A circular shaft transmits torque T. Each node has **one dof**, the angle of twist φ (rotation about the shaft axis). A 2-noded element has $\{q\} = [\varphi_1\ \varphi_2]^T$ - the formulation is identical to the 1-D bar element (Unit I) with the analogy :

Axial bar	Torsion of shaft
displacement u	angle of twist φ
axial force F	torque T
E	G (shear modulus)
A	J (polar moment of area)
AE/L	GJ/L

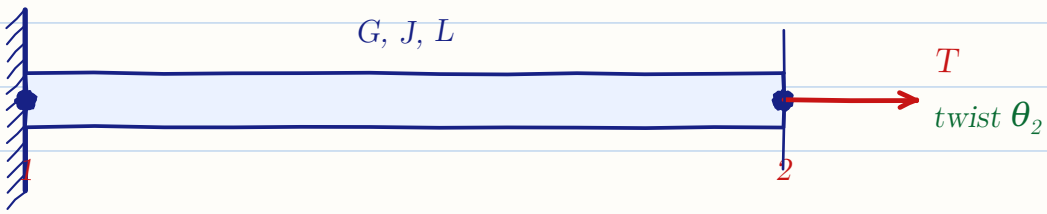


Fig 5.1 Shaft element

Basic relations : $T/J = G\gamma/r = G\, d\varphi/dx \Rightarrow T = G\, J\, d\varphi/dx$. Polar moment : $J = \pi d^4/32$ (solid), $J = \pi(D^4 - d^4)/32$ (hollow). Shear stress $\tau = G\, r\, d\varphi/dx = T\, r/J$; maximum at the outer surface.

Finite element formulation

$$\varphi = N_1\varphi_1 + N_2\varphi_2,\ N_1 = (1-\xi)/2,\ N_2 = (1+\xi)/2\ ;\ d\varphi/dx = (1/L)[-1\ 1]\{\varphi_1\ \varphi_2\}^T$$

Strain energy $U = \tfrac{1}{2} \int G\, J\, (d\varphi/dx)^2\, dx = \tfrac{1}{2}\{q\}^T[k]\{q\}$ and work of the nodal torques $\{T\}$:

$$[k] = GJ/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} \varphi_1 \\ \varphi_2 \end{Bmatrix} = [T_1\ T_2]^T$$

Assemble exactly as for bars : node i of an n-element shaft contributes GJ_{i-1}/L_{i-1} + GJ_i/L_i to K_{ii}. Boundary condition : $\varphi = 0$ at a fixed end (elimination/penalty). Torques are applied at nodes; for a distributed torque t per unit length : $\{f\} = tL/2\ [1\ 1]^T$.

Shear stress in an element : $\tau = G\ (d/2)(\varphi_2 - \varphi_1)/L$ (d = diameter of that element).

5.2 Solved Problems on Shafts

Problem 1 : stepped shaft with end torque

Shaft fixed at node 1; element 1 : $d = 40$ mm, $L = 500$ mm; element 2 : $d = 30$ mm, $L = 500$ mm; $G = 8\times10^4$ N/mm^2; torque $T = 10^6$ N · mm at the free end (node 3).

$J_1 = \pi(40)^4/32 = 251327$ mm^4, $J_2 = \pi(30)^4/32 = 79522$ mm^4. $GJ_1/L_1 = 40212386$, $GJ_2/L_2 = 12723450$ N · mm/rad.

$$10^6 \times \begin{bmatrix} 52.936 & -12.723 \\ -12.723 & 12.723 \end{bmatrix} \begin{Bmatrix} \varphi_2 \\ \varphi_3 \end{Bmatrix} = [0,\ 10^6]^T$$

Solution : $\varphi_2 = 0.02487$ rad, $\varphi_3 = 0.10346$ rad (= 5.928 degrees). Shear stresses : $\tau_1 = G(d_1/2)\varphi_2/L_1 = 79.58$ N/mm^2, $\tau_2 = G(d_2/2)(\varphi_3-\varphi_2)/L_2 = 188.63$ N/mm^2.

Check : $\tau = T\,r/J$: element 1 $\rightarrow 10^6\times20/251327 = 79.58$; element 2 $\rightarrow 10^6\times15/79522 = 188.63$ N/mm^2 (same torque flows through both elements).

Problem 2 : shaft fixed at both ends, torque at the middle

Uniform shaft, 2 elements of equal length $L/2$, torque T at node 2, nodes 1 and 3 fixed. Reduced equation : $(2GJ/(L/2)\ldots) = (GJ/L_e)\ [2]\ \varphi_2 = T$ with $L_e = L/2$:

$$\varphi_2 = T\,L_e/(2\,G\,J) = T\,L/(4\,G\,J) \text{ reactions } T_1 = T_3 = T/2$$

Torque in each half = $T/2$ with opposite sense; $\tau_{max} = T\,r/(2J)$.

Problem 3 : three-element shaft with two torques

Uniform shaft $d = 40$ mm, three elements of 300 mm (nodes 1-4), $G = 8\times10^4$ N/mm^2, fixed at node 1; torques 5×10^5 N · mm at node 2 and 3×10^5 N · mm at node 4. $GJ/L = 67020643$ N · mm/rad.

Node	1	2	3	4
φ (rad)	0.00000	0.01194	0.01641	0.02089

Shear stresses : $\tau_1 = 63.66$, $\tau_2 = 23.87$, $\tau_3 = 23.87$ N/mm^2. Reaction torque at node 1 = -800000 N · mm (= $-(5+3)\times10^5$; equilibrium OK).

5.3 Dynamic Analysis : Formulation of the FE Model

For a structure vibrating under time-dependent loads, inertia must be included. Using **Hamilton's principle / Lagrange's equation** with kinetic energy $T = \frac{1}{2} \int \rho \{\dot{u}\}^T \{\dot{u}\} dV$, strain energy $U = \frac{1}{2}\{Q\}^T[K]\{Q\}$, we obtain the equation of motion :

$$[M]\{\ddot{Q}\} + [C]\{\dot{Q}\} + [K]\{Q\} = \{F(t)\}$$

$[M]$ = mass matrix, $[C]$ = damping matrix (Rayleigh : $[C] = a[M] + b[K]$), $[K]$ = stiffness matrix, $\{Q\}$ = nodal displacements. **Free vibration** (no damping, no load) with $Q = \{\varphi\} \sin \omega t$:

$$([K] - \omega^2[M]) \{\varphi\} = \{0\} \Rightarrow \det([K] - \lambda[M]) = 0, \lambda = \omega^2$$

- Eigenvalues $\lambda_i = \omega_i^2$ give natural frequencies ω_i (rad/s), $f_i = \omega_i/2\pi$ (Hz). Number of eigenvalues = number of free dof.
- Eigenvectors $\{\varphi\}_i$ give the **mode shapes**; they are determined up to a scale factor. Normalisation : $\{\varphi\}_i^T[M]\{\varphi\}_i = 1$.
- **Orthogonality** : $\{\varphi\}_i^T[M]\{\varphi\}_j = 0$ and $\{\varphi\}_i^T[K]\{\varphi\}_j = 0$ for $i \neq j$; $\{\varphi\}_i^T[K]\{\varphi\}_i = \omega_i^2$.
- Rigid-body modes of an unrestrained structure have $\omega = 0$ (before applying supports, $[K]$ is singular).

Element mass matrices

From kinetic energy, with displacement $u = [N]\{q\}$:

$$[m] = \int_V \rho\, [N]^T[N]\, dV \text{ (**consistent** mass matrix, same shape functions as stiffness)}$$

$$\text{1-D bar : } [m] = \rho AL/6 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\text{Torsion shaft : } [m] = \rho JL/6 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Derivation (bar) : $[m] = \rho A \int_0^L [N_1\ N_2]^T[N_1\ N_2]\, dx = \rho A\ (L/2) \int_{-1}^{1} [\ N_1^2\ N_1N_2\ ;\ N_1N_2\ N_2^2\ ]\, d\xi$; $\int N_1^2 d\xi \cdot (L/2) = L/3$, $\int N_1N_2\, d\xi \cdot (L/2) = L/6 \Rightarrow \rho AL/6\ [2\ 1\ ;\ 1\ 2]$.

$$\textbf{Lumped} \text{ mass (half the mass at each node) : } [m] = \rho AL/2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Plane truss : } [m] = \rho AL/6 \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

$$\text{Beam : } [m] = \rho AL/420 \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}$$

Remark: *Total mass check : the sum of all entries of the translational part of [m] must equal the element mass ρAL (bar : $(2+1+1+2)\rho AL/6 = \rho AL$). Lumped matrices are diagonal, cheaper, and give a lower-bound (under-estimate) of frequencies, while consistent matrices give an upper bound.*

(lumped model of a bar / shaft)

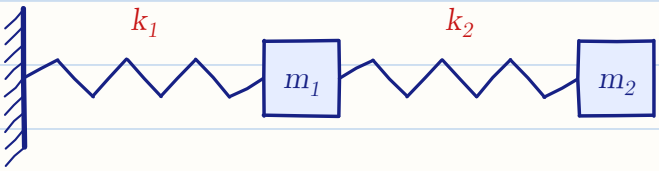


Fig 5.2 Lumped spring-mass equivalent of a bar

5.4 Eigenvalues and Eigenvectors : Bar and Shaft

Procedure

1. *Find [k] and [m] for each element; assemble [K], [M].*
2. *Apply boundary conditions (delete rows/columns of restrained dof) - **both** [K] and [M].*
3. *Solve the characteristic equation $\det([K] - \lambda[M]) = 0$ for $\lambda_i = \omega_i^2$.*
4. *Substitute each λ_i in $([K] - \lambda_i[M])\{\varphi\} = 0$, fix one component = 1 and find the others (mode shape).*
5. *Normalise the modes if required and check orthogonality.*

Problem 4 : fixed-free bar, one element

Uniform bar fixed at node 1, free at node 2 (L, A, E, ρ). Node 2 only : $K = AE/L$, $M = \rho AL/3$ (consistent) or $\rho AL/2$ (lumped).

Consistent : $\omega^2 = 3E/(\rho L^2) \Rightarrow \omega = 1.732\ \sqrt{(E/\rho)}/L$ Lumped : $\omega = 1.414\ \sqrt{(E/\rho)}/L$ Exact : $1.571\ \sqrt{(E/\rho)}/L$

For steel ($E = 2\times10^5$ N/mm^2, $\rho = 7.85\times10^{-9}$ N·s^2/mm^4), $L = 1000$ mm : $\sqrt{(E/\rho)}/L = 5047.5$ rad/s $\Rightarrow$ exact $\omega_1 = 1.5708\times5047.5 = 7929$ rad/s (1262 Hz); consistent 1-element = 8742 rad/s; lumped = 7137 rad/s.

Problem 5 : fixed-free bar, two elements (consistent mass)

Let $L_e = L/2$. Reduced matrices (nodes 2, 3) : $[K] = (AE/L_e)\ [2\ -1\ ;\ -1\ 1]$, $[M] = (\rho AL_e/6)\ [4\ 1\ ;\ 1\ 2]$.

$\det([K] - \lambda[M]) = 0$ with $\Lambda = \lambda\ \rho\ L_e^2/(6E)$: $(2 - 4\Lambda)(1 - 2\Lambda) - (-1 - \Lambda)^2 = 0$ $\Rightarrow 7\Lambda^2 - 10\Lambda + 1 = 0 \Rightarrow \Lambda = 0.1082,\ 1.3203.$

$\omega_1 = \sqrt{(6\times0.1082)}\ \sqrt{(E/\rho)}/L_e = 0.806\ \sqrt{(E/\rho)}/L_e = 1.611\ \sqrt{(E/\rho)}/L$ $\omega_2 = 2.815\ \sqrt{(E/\rho)}/L_e = 5.629\ \sqrt{(E/\rho)}/L$

Mode	Exact ($\pi(2n-1)/2$)	1 elem (cons.)	2 elem (cons.)	Error (2 elem)
1	1.5708	1.7321	1.6113	2.6 %
2	4.7124	–	5.6293	19.5 %

Mode shapes (normalised to the tip = 1) : substituting Λ_1 in $(2-4\Lambda)\varphi_2 - (1+\Lambda)\varphi_3 = 0$: $\{\varphi\}_1 = [0.7071\ ,\ 1]^T$ (node 2, node 3), $\{\varphi\}_2 = [-0.7071\ ,\ 1]^T$. The first mode has no sign change; the second has one node (sign change) - matches $\sin(\pi x/2L)$ and $\sin(3\pi x/2L)$.

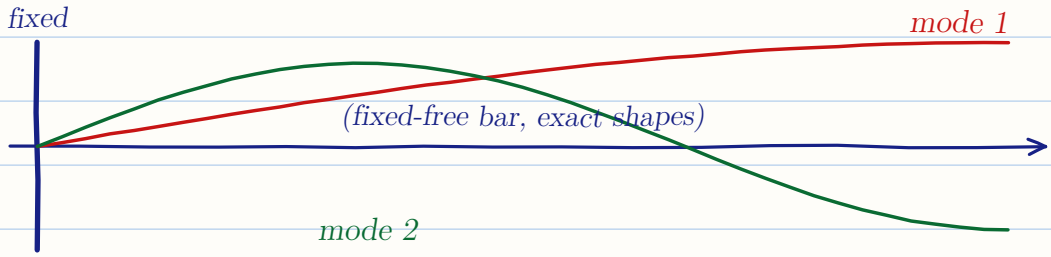


Fig 5.3 First two modes of the fixed-free bar

Orthogonality check : $\{\varphi\}_1^T[M]\{\varphi\}_2 = 0.00000 \approx 0$ OK.

5.5 Solved Problems : Shaft Vibration

Problem 6 : torsional vibration of a fixed-free shaft

Shaft $L = 1000$ mm, $d = 50$ mm, $G = 8\times10^4$ N/mm^2, $\rho = 7.85\times10^{-9}$ N $\cdot$ s^2/mm^4.

Two elements. The matrices are the bar matrices with ($AE \to GJ$, $\rho A \to \rho J$), so :

$$[K] = GJ/L_e \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$[M] = \rho J L_e/6 \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$$

$\omega_1 = 1.611\ \sqrt{(G/\rho)}/L = 1.611 \times 3192 = 5144$ rad/s (818.7 Hz); $\omega_2 = 5.629 \times 3192 = 17971$ rad/s. Exact : $\omega_1 = (\pi/2)\sqrt{(G/\rho)}/L = 5015$ rad/s.

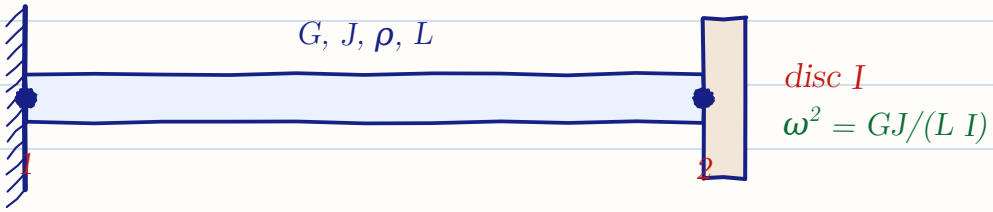


Fig 5.4 Shaft with a disc at the free end

Problem 7 : shaft carrying a disc (massless shaft, one element)

Shaft $d = 30$ mm, $L = 500$ mm, $G = 8\times10^4$ N/mm^2; disc of polar mass moment of inertia $I = 0.05$ kg $\cdot$ m^2 $= 50$ N $\cdot$ mm $\cdot$ s^2. With one dof φ_2 : $K = GJ/L = 12723450$ N $\cdot$ mm/rad, $M = I = 50$:

$$\omega^2 = GJ/(L\ I) = 254469 \Rightarrow \omega = 504.4 \text{ rad/s } (f = 80.3 \text{ Hz})$$

If shaft inertia is to be included, add $\rho JL/3$ (consistent) to the disc inertia at the free node : $M = I + \rho JL/3$ (small here : $\rho JL/3 = 0.10404$ N $\cdot$ mm $\cdot$ s^2), so ω decreases very slightly.

Problem 8 : free-free bar (unrestrained) - rigid body mode

One element with both nodes free : $\det([K] - \lambda[M]) = 0$ with $[K] = (AE/L)[1\ -1\ ;\ -1\ 1]$, $[M] = (\rho AL/6)[2\ 1\ ;\ 1\ 2]$:

$$\lambda_1 = 0 \text{ (rigid-body translation) } \lambda_2 = 12E/(\rho L^2) \Rightarrow \omega_2 = 3.464\ \sqrt{(E/\rho)}/L \text{ (exact } \pi = 3.142)$$

Eigenvectors : $\{\varphi\}_1 = [1,\ 1]^T$ (both nodes move together), $\{\varphi\}_2 = [1,\ -1]^T$ (nodes move oppositely). λ_2 computed numerically $= 12.0000 \times E/(\rho L^2)$ OK.

Problem 9 : cantilever beam frequency (1 element)

With the beam matrices (Unit II and above), deleting v_1, θ_1 : $(EI/L^3)[12\ -6L\ ;\ -6L\ 4L^2]$, $(\rho AL/420)[156\ -22L\ ;\ -22L\ 4L^2]$. Solving gives

$\omega_1 = 3.533\ \sqrt{(EI/\rho AL^4)} = 3.533\ \sqrt{(EI/\rho A)}/L^2$ (exact 3.516) $\omega_2 = 34.81\ \sqrt{(EI/\rho AL^4)}$ (exact 22.03)

Observation: A single cubic Hermite element gives the first frequency within 0.5 % but the second mode is poor (about 58 % high). More elements are required for higher modes: rule of thumb - use about 2 elements per half-wave of the highest mode of interest.

5.6 Derivation of the Dynamic Equations

Lagrange's equation for a conservative system with generalised coordinates Q_i :

$d/dt\ (\partial T/\partial \dot{Q}_i) - \partial T/\partial Q_i + \partial U/\partial Q_i = F_i(t)$.

1. Kinetic energy of an element : $T_e = \tfrac{1}{2} \int \rho\ \{\dot{u}\}^T\{\dot{u}\}\ dV$, with $\{\dot{u}\} = [N]\{\dot{q}\} \Rightarrow$ $T_e = \tfrac{1}{2}\ \{\dot{q}\}^T\ (\int \rho\ [N]^T[N]\ dV\ )\ \{\dot{q}\} = \tfrac{1}{2}\ \{\dot{q}\}^T[m]\{\dot{q}\}$.
2. Strain energy $U_e = \tfrac{1}{2}\ \{q\}^T[k]\{q\}$. Sum over elements : $T = \tfrac{1}{2}\ \{\dot{Q}\}^T[M]\{\dot{Q}\}$, $U = \tfrac{1}{2}\ \{Q\}^T[K]\{Q\}$.
3. Substitute in Lagrange's equation : $\partial T/\partial \dot{Q} = [M]\{\dot{Q}\}$, d/dt gives $[M]\{\ddot{Q}\}$; $\partial U/\partial Q = [K]\{Q\}$. Hence $[M]\{\ddot{Q}\} + [K]\{Q\} = \{F(t)\}$.
4. Damping adds $[C]\{\dot{Q}\}$ (dissipation function $\tfrac{1}{2}\{\dot{Q}\}^T[C]\{\dot{Q}\}$).

Consistent mass matrix of the beam element (outline)

$[m] = \rho A \int_0^L [H]^T[H]\ dx = \rho A\ (L/2) \int_{-1}^{1} [H]^T[H]\ d\xi$, with the Hermite cubics of Unit II. Example : $m_{11} = \rho A\ (L/2)(1/16) \int\ (1-\xi)^4(2+\xi)^2\ d\xi = \rho A\ L\ (156/420)$; $m_{22} = \rho A\ (L/2)(L^2/64) \int\ (1-\xi)^4(1+\xi)^2\ d\xi = \rho A\ L^3(4/420)$.

Forced vibration (harmonic) in brief

For $F(t) = F_0 \sin \Omega t$ the steady response is $\{Q\} = \{Q_0\} \sin \Omega t$ with $(\ [K] - \Omega^2[M]\ )\{Q_0\} = \{F_0\}$. When Ω approaches a natural frequency ω_i the matrix becomes singular - **resonance**. With damping the amplitude of mode i is limited by $1/(2\zeta_i)$.

5.7 More Solved Problems

Problem 10 : convergence of bar frequencies with mesh refinement

Fixed-free bar. Frequencies are listed as the coefficient C in $\omega = C\sqrt{(E/\rho)}/L$. Consistent mass gives upper bounds, lumped mass gives lower bounds, and both converge to the exact values :

Elements	Consistent ω_1	Consistent ω_2	Consistent ω_3	Lumped ω_1	Lumped ω_2
1	1.7321	–	–	1.4142	–
2	1.6114	5.6293	–	1.5307	3.6955
3	1.5888	5.1962	9.4266	1.5529	4.2426
4	1.5809	4.9872	9.0594	1.5607	4.4446
6	1.5753	4.8342	8.4188	1.5663	4.5922
Exact	1.5708	4.7124	7.8540	1.5708	4.7124

Observation : the lower modes converge fast; the 3rd mode needs at least 4-6 elements. The **average** of consistent and lumped results is often very close to the exact value.

Problem 11 : fixed-fixed bar, three elements

Interior nodes 2, 3 free (4 nodes, ends fixed); $[K] = (AE/L_e)[2\ -1\ ;\ -1\ 2]$, $[M] = (\rho AL_e/6)[4\ 1\ ;\ 1\ 4] \Rightarrow \det = 0$ gives $(2-4\Lambda)^2 - (1+\Lambda)^2 = 0 \Rightarrow \Lambda = 1/5,\ 1 \Rightarrow \omega = \sqrt{(6\Lambda)}/L_e$. With $L_e = L/3$:

Mode	FEM (3 elem, consistent)	Exact nπ	Error %
1	3.2863	3.1416	4.61
2	7.3485	6.2832	16.95

Mode shapes : mode 1 $\{\varphi\} = [1,\ 1]^T$ (symmetric, both interior nodes move the same way); mode 2 $\{\varphi\} = [1,\ -1]^T$ (antisymmetric, a node at the centre).

Problem 12 : two-disc shaft (torsional vibration)

The stepped shaft of Problem 1 ($k_1 = 40212386$, $k_2 = 12723450$ N · mm/rad) is fixed at node 1 and carries discs $I_2 = 100$ at node 2 and $I_3 = 50$ N · mm · s^2 at node 3 (shaft mass neglected, so $[M]$ is diagonal).

$$[K]\ (\times 10^6) = \begin{bmatrix} 52.936 & -12.723 \\ -12.723 & 12.723 \end{bmatrix}$$

Frequency equation : $\det([K] - \omega^2[M]) = 0 \Rightarrow \omega_1 = 406.8$ rad/s, $\omega_2 = 786.3$ rad/s

(f = 64.7, 125.2 Hz).

Mode shapes (disc 2 = 1) : $\{\varphi\}_1 = [1,\ 2.860]^T$ (both discs rotate in the same sense), $\{\varphi\}_2 = [1,\ -1.414]^T$ (opposite senses - one node on the shaft between the discs).

Problem 13 : two-dof spring-mass (lumped bar) - hand solution

Fixed-free bar of two equal elements with lumped masses m at node 2 and m at node 3 (end mass taken = m for simplicity); $k = AE/L_e$. $[K] = k[2\ -1\ ;\ -1\ 1]$, $[M] = m\ [1\ 0\ ;\ 0\ 1]$.

$(2k - \lambda m)(k - \lambda m) - k^2 = 0 \Rightarrow \lambda^2 m^2 - 3km\lambda + k^2 = 0 \Rightarrow \lambda = (k/m)(3 \mp \sqrt{5})/2 = 0.382\ k/m\ ,\ 2.618\ k/m$

$\omega_1 = 0.618\ \sqrt{(k/m)}$, $\omega_2 = 1.618\ \sqrt{(k/m)}$. Mode shapes : from $(2k - \lambda m)\varphi_2 - k\varphi_3 = 0$: $\lambda_1 \Rightarrow \varphi_3/\varphi_2 = 2 - 0.382 = 1.618 \Rightarrow \{\varphi\}_1 = [1,\ 1.618]^T$; $\lambda_2 \Rightarrow \varphi_3/\varphi_2 = 2 - 2.618 = -0.618 \Rightarrow \{\varphi\}_2 = [1,\ -0.618]^T$.

Orthogonality with $[M] = mI$: $1\times1 + 1.618\times(-0.618) = 1 - 1.000 = 0$ OK.

Natural frequency checklist

- Convert units first : $\rho = 7850\ kg/m^3 = 7.85\times10^{-9}\ N\cdot s^2/mm^4$ (in N-mm-s system) ; $E = 200\ GPa = 2\times10^5\ N/mm^2$; $G = 80\ GPa = 8\times10^4\ N/mm^2$.
- Use the **same** BC for $[K]$ and $[M]$. ω in rad/s ; $f = \omega/2\pi$; $T = 1/f$.
- Number of modes = number of free dof; FE frequencies from a consistent mass are upper bounds of the exact ones.

Time response and damping (brief)

- Modal superposition : $\{Q\} = \Sigma\ \{\varphi\}_i\ \eta_i(t)$; the equations decouple to $\ddot{\eta}_i + 2\zeta_i\omega_i\ \dot{\eta}_i + \omega_i^2\eta_i = \{\varphi\}_i^T\{F(t)\}$. Direct methods : central difference, Newmark-β.
- Numerical eigen-solvers : characteristic polynomial (small systems), inverse (power) iteration, Jacobi, subspace iteration, Lanczos (large systems).
- Rayleigh quotient $\omega^2 \approx \{\varphi\}^T[K]\{\varphi\}/\{\varphi\}^T[M]\{\varphi\}$ gives an upper bound to ω_1^2 for any trial vector.

Summary of mass and stiffness matrices

Element	[k]	[m] consistent
Bar	AE/L [1 −1 ; −1 1]	$\rho AL/6$ [2 1 ; 1 2]
Torsion	GJ/L [1 −1 ; −1 1]	$\rho JL/6$ [2 1 ; 1 2]
Beam	EI/L^3[12 6L ...]	$\rho AL/420$ [156 22L ...]
CST (plane)	$tA\ B^TDB$	$\rho tA/12$ [2 0 1 0 1 0 ; ...]

5.8 Further Examples and Theory

Problem 14 : cantilever beam, convergence with elements

Cantilever beam of uniform section; the results are the coefficient C in $\omega = C\sqrt{(EI/\rho A)/L^2}$ (consistent mass, Hermite elements) :

Elements	ω_1	ω_2	ω_3
1	3.533	34.81	–
2	3.518	22.22	75.16
4	3.516	22.06	62.17
Exact	3.516	22.03	61.70

Two elements already give ω_1 to 4 digits; higher modes converge more slowly - a typical property of FE eigenvalue solutions.

Problem 15 : Rayleigh quotient estimate

For the fixed-free bar take the trial shape $u = 2x/L - x^2/L^2$ (satisfies $u(0) = 0$, and $u'(L) = 0$).

$$\omega^2 \approx \int AE\,(u')^2 dx \,/\, \int \rho A\, u^2\, dx = [4AE/(3L)] \,/\, [\rho AL\ (8/15)] = 2.5\ E/(\rho L^2) \Rightarrow \omega = 1.581\sqrt{(E/\rho)}/L$$

Exact = 1.5708; the estimate is an **upper bound** (error 0.7 %). A trial $u = x/L$ gives $\omega = 1.732$, i.e. the one-element FE value. Rayleigh quotient is stationary, so a first-order error in the shape produces only a second-order error in ω^2.

Problem 16 : bar with an end mass

Steel bar ($A = 100\ mm^2$, $L = 500\ mm$, $E = 2\times10^5$) fixed at one end, carrying a lumped mass $m = 2\ kg = 2\times10^{-3}\ N\cdot s^2/mm$ at the free end; bar mass neglected, one element : $K = AE/L = 4\times10^4\ N/mm$.

$$\omega = \sqrt{(K/m)} = \sqrt{(4\times10^4/2\times10^{-3})} = 4472.1\ rad/s\ (f = 711.8\ Hz)$$

Adding the bar mass (consistent) : $M = m + \rho AL/3 = 2\times10^{-3} + 0.000131 = 0.002131 \Rightarrow \omega = 4332.7$ rad/s (slightly lower - the bar mass is only 19.63 % of the end mass).

Mass matrices of 2-D elements (for reference)

$$CST:\ [m] = \rho tA/12 \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 \end{bmatrix}$$

- Lumped CST mass : $\rho tA/3$ at each translational dof. Q4 : $[m] = \rho\ t \int\int [N]^T[N]\ \det J\ d\xi\ d\eta$ (2×2 Gauss). Axisymmetric ring : replace tA by $2\pi\ \bar{r}\ A$.
- Row-sum lumping : replace each row of the consistent matrix by its sum on the diagonal. For the bar : $\rho AL/6(2+1) = \rho AL/2$ OK.
- HRZ lumping is used for higher-order elements to avoid negative masses (Q8 corner nodes).

Damping

Rayleigh damping $[C] = \alpha[M] + \beta[K]$. Modal damping ratio $\zeta_i = \alpha/(2\omega_i) + \beta\omega_i/2$. Given ζ at two frequencies ω_1, ω_2 : $\alpha = 2\omega_1\omega_2(\zeta_1\omega_2 - \zeta_2\omega_1)/({\omega_2}^2 - {\omega_1}^2)$, $\beta = 2(\zeta_2\omega_2 - \zeta_1\omega_1)/({\omega_2}^2 - {\omega_1}^2)$.

Direct time integration (reference)

Method	Formula	Nature
Central difference	$Q_{n+1} = 2Q_n - Q_{n-1} + \Delta t^2\ M^{-1}(F - KQ)_n$	explicit, conditionally stable ($\Delta t < 2/\omega_{max}$)
Newmark-β (β=1/4, γ=1/2)	average acceleration	implicit, unconditionally stable

Short answer questions

Q1. What is a consistent mass matrix ? - One derived with the same shape functions used for stiffness : $[m] = \int \rho [N]^T [N] dV$.

Q2. Write the equation of free vibration. - $([K] - \omega^2 [M])\{\varphi\} = 0$.

Q3. What is the analogy between bar and shaft ? - $u \rightarrow \varphi$, $AE \rightarrow GJ$, $\rho A \rightarrow \rho J$.

Q4. Why is [M] of a bar not diagonal in the consistent form ? - Because $N_1 N_2 \neq 0$ couples the two nodes.

Q5. Units of ρ in the N-mm system ? - $N \cdot s^2/mm^4$ (steel 7.85×10^{-9}).

Q6. What are the physical meanings of eigenvalue and eigenvector ? - ω^2 and mode shape (relative amplitudes of dof in free vibration).

Expected questions

- Derive the stiffness matrix for a shaft element; solve stepped/uniform shaft for twist and shear stress.
- Derive consistent and lumped mass matrices of a bar; write the eigenvalue problem.
- Find natural frequencies and mode shapes of a fixed-free bar / shaft with 1 or 2 elements; check orthogonality.
- Write the equation of motion for the dynamic analysis; what are modal analysis and damping ?